

Topic B12

Respiration

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B12.1	Aerobic respiration
B12.2	Anaerobic respiration
B12.3	Oxygen debt
B12.4	Practical investigations

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Aerobic respiration

Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules and release energy. Glucose and oxygen form carbon dioxide and water. Energy supports muscle contraction, active transport, protein synthesis, cell division and temperature maintenance.

Examiner insight

glucose + oxygen → carbon dioxide + water, with energy transferred.

2. Anaerobic respiration

Anaerobic respiration releases less energy per glucose because breakdown is incomplete. In human muscles, glucose forms lactic acid. In yeast, glucose forms ethanol and carbon dioxide; this fermentation is used in bread and alcoholic-drink production.

Examiner insight

Do not include oxygen in an anaerobic equation.

3. Exercise and oxygen debt

During vigorous exercise, oxygen delivery may not meet demand and muscles respire anaerobically. Lactic acid accumulates and contributes to fatigue. After exercise, elevated heart and ventilation rates supply oxygen to oxidise lactic acid or convert it in the liver; this is excess post-exercise oxygen consumption.

Examiner insight

Avoid saying lactic acid itself “is oxygen debt”; the debt is the extra oxygen required after exercise.

4. Measuring respiration

Germinating seeds release carbon dioxide and thermal energy. Respirometers measure oxygen uptake when carbon dioxide is absorbed. Controls use dead seeds or glass beads and constant temperature; a water bath reduces pressure changes unrelated to respiration.

Examiner insight

Movement in a respirometer reflects gas-volume or pressure change, not organisms physically pulling liquid.

Worked application

Situation: A sprinter continues breathing rapidly after finishing.

Prediction: Extra oxygen is required.

Reason: Oxygen supports removal or oxidation of lactic acid and restoration of normal conditions.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B12

Respiration

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A - Multiple choice

Choose the best answer for each question.

1. Which statement best defines respiration? [1]
 - A chemical reactions in cells that break down nutrient molecules and release energy
 - B energy-releasing breakdown of nutrients using oxygen
 - C energy release without oxygen
 - D extra oxygen required after exercise to deal with lactic acid
2. Which statement best defines aerobic respiration? [1]
 - A energy release without oxygen
 - B energy-releasing breakdown of nutrients using oxygen
 - C extra oxygen required after exercise to deal with lactic acid
 - D anaerobic respiration by microorganisms such as yeast
3. Which statement best defines anaerobic respiration? [1]
 - A extra oxygen required after exercise to deal with lactic acid
 - B anaerobic respiration by microorganisms such as yeast
 - C energy release without oxygen
 - D chemical reactions in cells that break down nutrient molecules and release energy
4. Which statement best defines oxygen debt? [1]
 - A anaerobic respiration by microorganisms such as yeast
 - B chemical reactions in cells that break down nutrient molecules and release energy
 - C energy-releasing breakdown of nutrients using oxygen
 - D extra oxygen required after exercise to deal with lactic acid
5. Which statement best defines fermentation? [1]
 - A anaerobic respiration by microorganisms such as yeast
 - B chemical reactions in cells that break down nutrient molecules and release energy
 - C energy-releasing breakdown of nutrients using oxygen
 - D energy release without oxygen
6. Which statement about aerobic respiration is correct? [1]
 - A Movement of air into lungs.
 - B Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules and release energy.
 - C Respiration that forms lactic acid.
 - D Photosynthesis in darkness.

7. Which statement about anaerobic respiration is correct? [1]
- A Respiration that forms lactic acid.
 - B Photosynthesis in darkness.
 - C Anaerobic respiration releases less energy per glucose because breakdown is incomplete.
 - D Oxygen stored permanently in muscle.
8. Which statement about exercise and oxygen debt is correct? [1]
- A Photosynthesis in darkness.
 - B Oxygen stored permanently in muscle.
 - C Aerobic breakdown producing only water.
 - D During vigorous exercise, oxygen delivery may not meet demand and muscles respire anaerobically.
9. Which statement about measuring respiration is correct? [1]
- A Germinating seeds release carbon dioxide and thermal energy.
 - B Oxygen stored permanently in muscle.
 - C Aerobic breakdown producing only water.
 - D Movement of air into lungs.
10. Which prediction is most appropriate when a sprinter continues breathing rapidly after finishing? [1]
- A Movement of air into lungs.
 - B Photosynthesis in darkness.
 - C Extra oxygen is required.
 - D Aerobic breakdown producing only water.

Section B · Structured questions

11 Biologists use precise vocabulary when studying respiration.

(a) Define respiration. [2]

(b) Define aerobic respiration. [2]

(c) Distinguish anaerobic respiration from the incorrect idea: “photosynthesis in darkness”. [2]

(d) State two important features of aerobic respiration. [2]

[Total: 8]

12 A sprinter continues breathing rapidly after finishing.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records relative oxygen uptake under three conditions.

condition	relative oxygen uptake
10 °C	1
25 °C	5
40 °C	8

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to use a respirometer to test temperature on germinating seeds.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of respiration to an unfamiliar context.

(a) Define oxygen debt. [2]

(b) Explain the biological importance of exercise and oxygen debt. [3]

(c) Correct this statement: “aerobic breakdown producing only water”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. Chemical reactions in cells that break down nutrient molecules and release energy.
- 2 B. Energy-releasing breakdown of nutrients using oxygen.
- 3 C. Energy release without oxygen.
- 4 D. Extra oxygen required after exercise to deal with lactic acid.
- 5 A. Anaerobic respiration by microorganisms such as yeast.
- 6 B. glucose + oxygen → carbon dioxide + water, with energy transferred.
- 7 C. Do not include oxygen in an anaerobic equation.
- 8 D. Avoid saying lactic acid itself “is oxygen debt”; the debt is the extra oxygen required after exercise.
- 9 A. Movement in a respirometer reflects gas-volume or pressure change, not organisms physically pulling liquid.
- 10 C. Oxygen supports removal or oxidation of lactic acid and restoration of normal conditions.

Solutions · Structured questions

11. (a) chemical reactions in cells that break down nutrient molecules and release energy [2]. (b) energy-releasing breakdown of nutrients using oxygen [2]. (c) energy release without oxygen; not photosynthesis in darkness [2]. (d) Credit any two accurate features from the guide: Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules and release energy. Glucose and oxygen form carbon dioxide and water. Energy supports muscle contraction, active transport, protein synthesis, cell division and temperature maintenance. [2].

12. (a) Extra oxygen is required [2]. (b) Oxygen supports removal or oxidation of lactic acid and restoration of normal conditions [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $8 - 1 = 7$ [2]. (b) The values change from 1 to 8, with the highest value 8. [2]. (c) Higher temperature increases enzyme-controlled respiration until damaging temperatures are reached [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Temperature [1]. (b) Movement of coloured liquid per time [1]. (c) Same seed mass, time, apparatus and carbon dioxide absorber [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Use replicate respirometers and thermostatic water baths [1].

15. (a) extra oxygen required after exercise to deal with lactic acid [2]. (b) Award linked points from: During vigorous exercise, oxygen delivery may not meet demand and muscles respire anaerobically. Lactic acid accumulates and contributes to fatigue. After exercise, elevated heart and ventilation rates supply oxygen to oxidise lactic acid or convert it in the liver; this is excess post-exercise oxygen consumption. [3]. (c) Correct statement: fermentation means anaerobic respiration by microorganisms such as yeast [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.